🍊 Orange Juice Fair Pricing

Classical Monte Carlo vs Quantum Amplitude Estimation

A Simulation-Based Comparison for Fair Pricing Decisions

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📋 Executive Summary

Metric	Classical MC	Quantum AE	Improvement
Error at N = 100	± \$2.20	± \$0.22	10×
Error at N = 10,000	± \$0.18	± \$0.006	31.6×
Price stability (N=50)	σ = \$0.15	σ = \$0.07	2×
Price stability (N=1000)	σ = \$0.07	σ ≈ \$0.00	∞
Decision clarity	Ambiguous	Clear ✅	Qualitative leap

💡 Bottom Line:

Quantum AE enables more precise, stable, and fair pricing decisions
with the same computational budget — turning ambiguous choices into confident ones.

🎯 Problem Statement

The Business Question

A juice vendor must choose a selling price that maximizes expected profit under uncertain demand.

Why "Fair" Pricing Matters

⚠️ Overpricing → lose customers, reduced demand

⚠️ Underpricing → leave profit on the table

✅ Fair pricing → reflects true demand dynamics

The Core Challenge

Estimating expected profit accurately under demand uncertainty. The precision of this estimate

directly determines the quality of the pricing

Pricing = Expectation Estimation

Goal: estimate $E[\text{Profit}(p)]$

Profit = (price − cost) × Demand

Demand is uncertain:

$D(p) = D_0 \cdot (1 - \alpha \cdot (p - p\_ref)) \cdot (1+\varepsilon)$

$\varepsilon \sim \text{Uniform}(-15\%, +15\%)$

Two approaches to estimate E[Profit]:

🎲 **Classical Monte Carlo**

⚛️ **Quantum Amplitude Estimation**

🔧 Model Parameters

Parameter	Symbol	Value	Description
Cost per cup	c	$2.50	Production cost
Reference demand	D_ref	100 cups	Demand at reference price
Reference price	p_ref	$5.00	Baseline price
Price sensitivity	α	0.30	Demand drop per $1 increase
Demand noise	ε	±15%	Random demand fluctuation

Mathematical Formulation

Demand: $D(p) = 100 \cdot (1 - 0.3 \cdot (p - 5.0)) \cdot (1 + \varepsilon)$

Profit: $\Pi(p) = (p - 2.5) \cdot D(p)$

Optimal: $p^* = (1/\alpha + p_{ref} + c) / 2 = \$5.42 \rightarrow$ Simulated: $5.40

Expected profit at optimal: $255.20

🎲 Classical Monte Carlo

Algorithm

For each candidate price p:

1. Draw N random demand scenarios
   $\varepsilon_i \sim \text{Uniform}(-0.15, +0.15)$
2. Compute profit for each scenario
   $\Pi_i = (p - c) \cdot D(p, \varepsilon_i)$
3. Average all samples
   $\hat{\Pi}(p) = (1/N) \cdot \Sigma\, \Pi_i$

Error Scaling

$$\text{Error} \sim \sigma / \sqrt{N} = O(1/\sqrt{N})$$

The Problem: Quadratically Expensive

2× more precise → 4× more samples
10× more precise → 100× more samples
100× more precise → 10,000× more samples

⚠️ At N = 100:

Profit estimate: \$255 ± \$2.20

That's almost 1% uncertainty

Quantum Amplitude Estimation (QAE)

Core Idea

Instead of sampling independently, QAE:

1. Encodes all demand scenarios into a quantum superposition
2. Computes profit and encodes it into the amplitude of a flag qubit
3. Uses quantum interference (Grover-like iterations) to extract the expected value

$A|0\rangle = \sqrt{(1-a)}|\psi_0\rangle|0\rangle + \sqrt{a}|\psi_1\rangle|1\rangle$
where $a = E[\text{normalized profit}]$

Error Scaling

Error ~ O(1/N)

The Advantage: Linear Scaling

2× more precise → 2× more queries
10× more precise → 10× more queries
100× more precise → 100× more queries

✅ At N = 100:

Profit estimate: $255 ± $0.22

10× more precise than MC!

Expected Profit vs Price

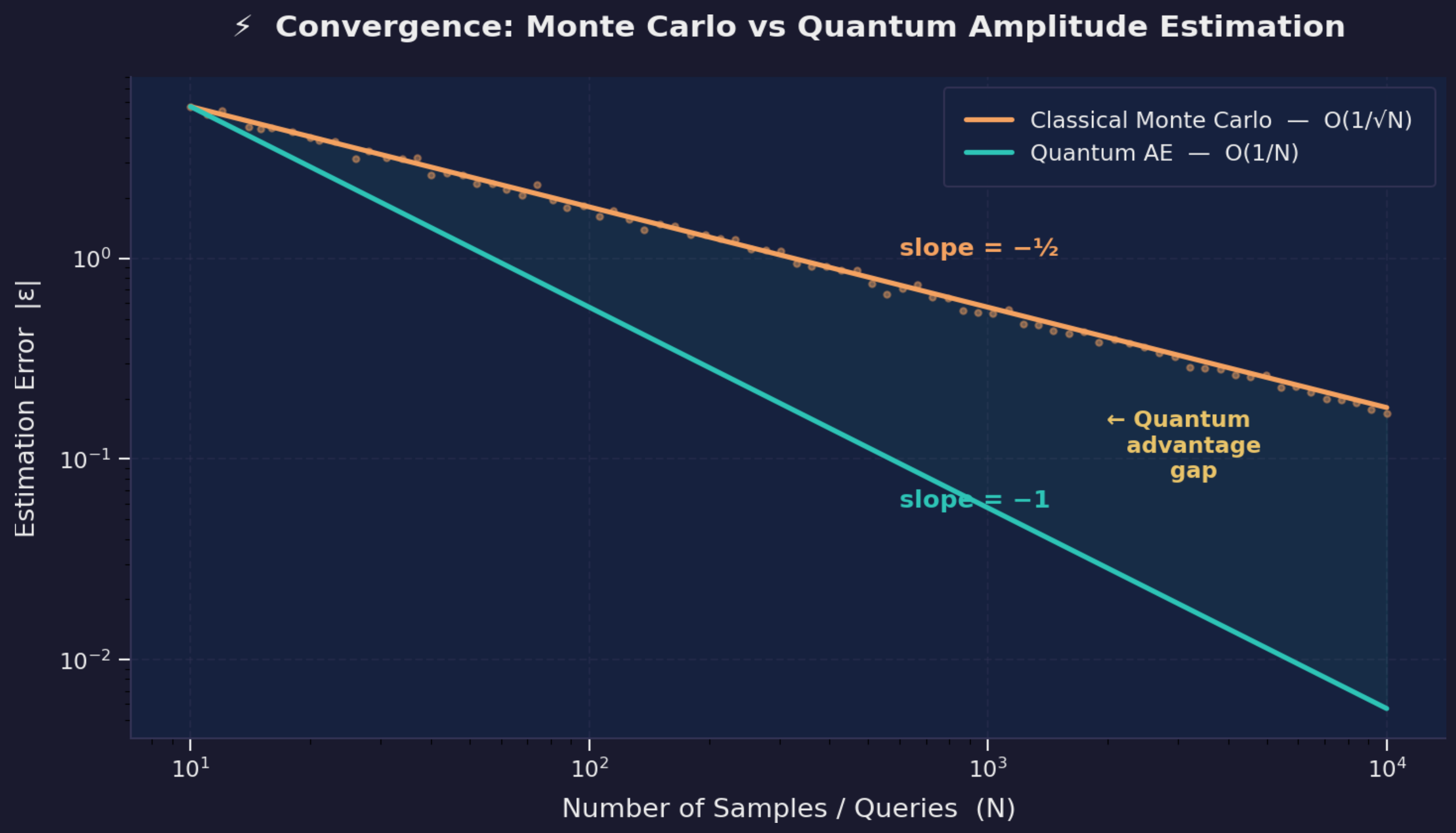
Key Insights

Optimal price: $5.40 → Expected profit: $255.20

The flat-top curve means small estimation errors can shift the perceived optimal price significantly

This is exactly where QAE's precision advantage matters most

Convergence: MC vs QAE



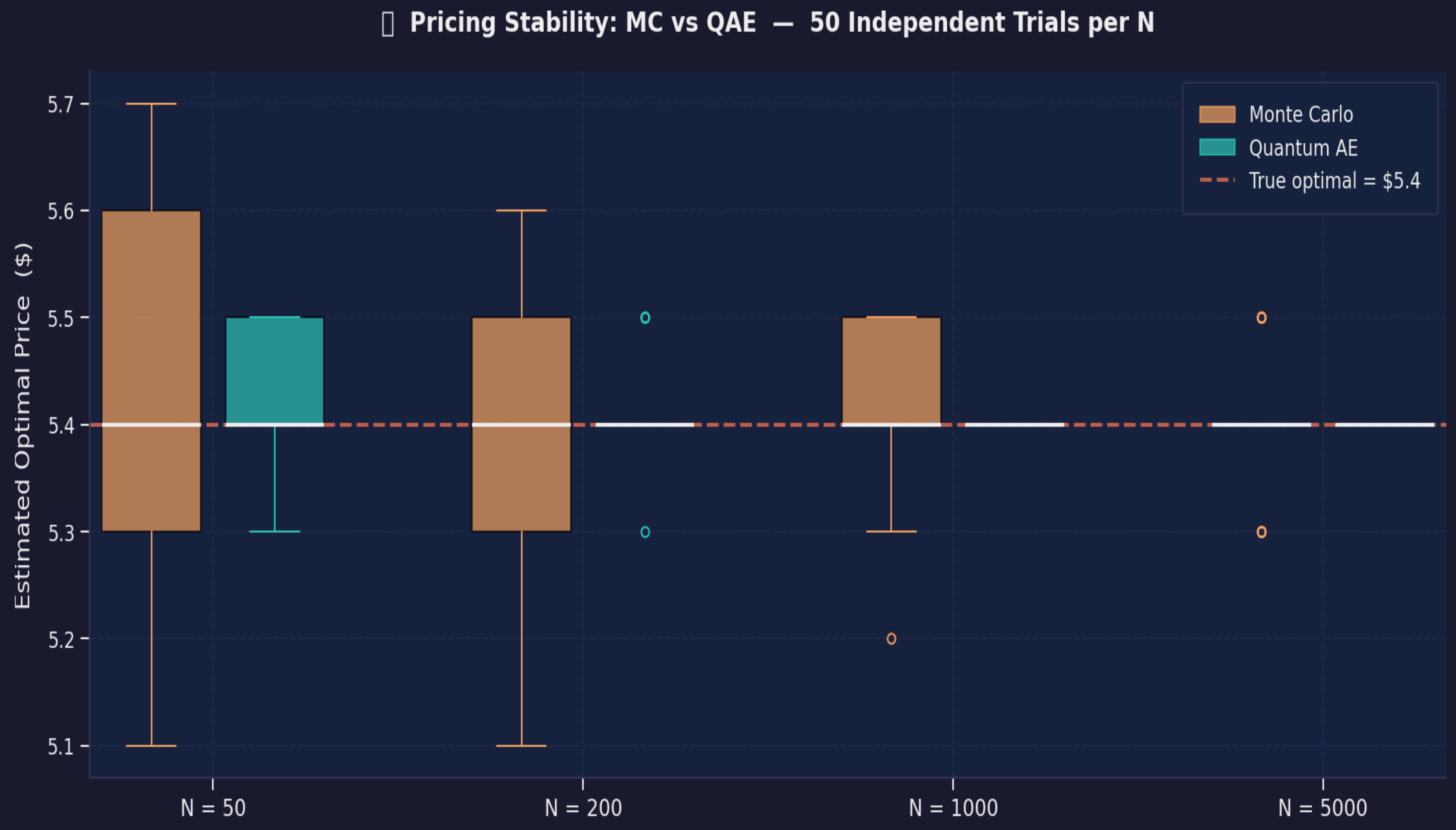
Key Insights

Orange (MC): slope −½ → O(1/√N) | Teal (QAE): slope −1 → O(1/N)

The shaded gap = quantum advantage — it grows with N

At N=10,000: QAE achieves 31.6× lower error than classical MC

Pricing Stability: 50 Independent Trials



Key Insights

At N=50: MC shows σ=$0.15 spread — the "optimal" price jumps around

QAE stabilizes much faster — near-deterministic even at low N

Stable pricing = consistent decisions = fairer outcomes

🔥 Profit Landscape: Price × Uncertainty

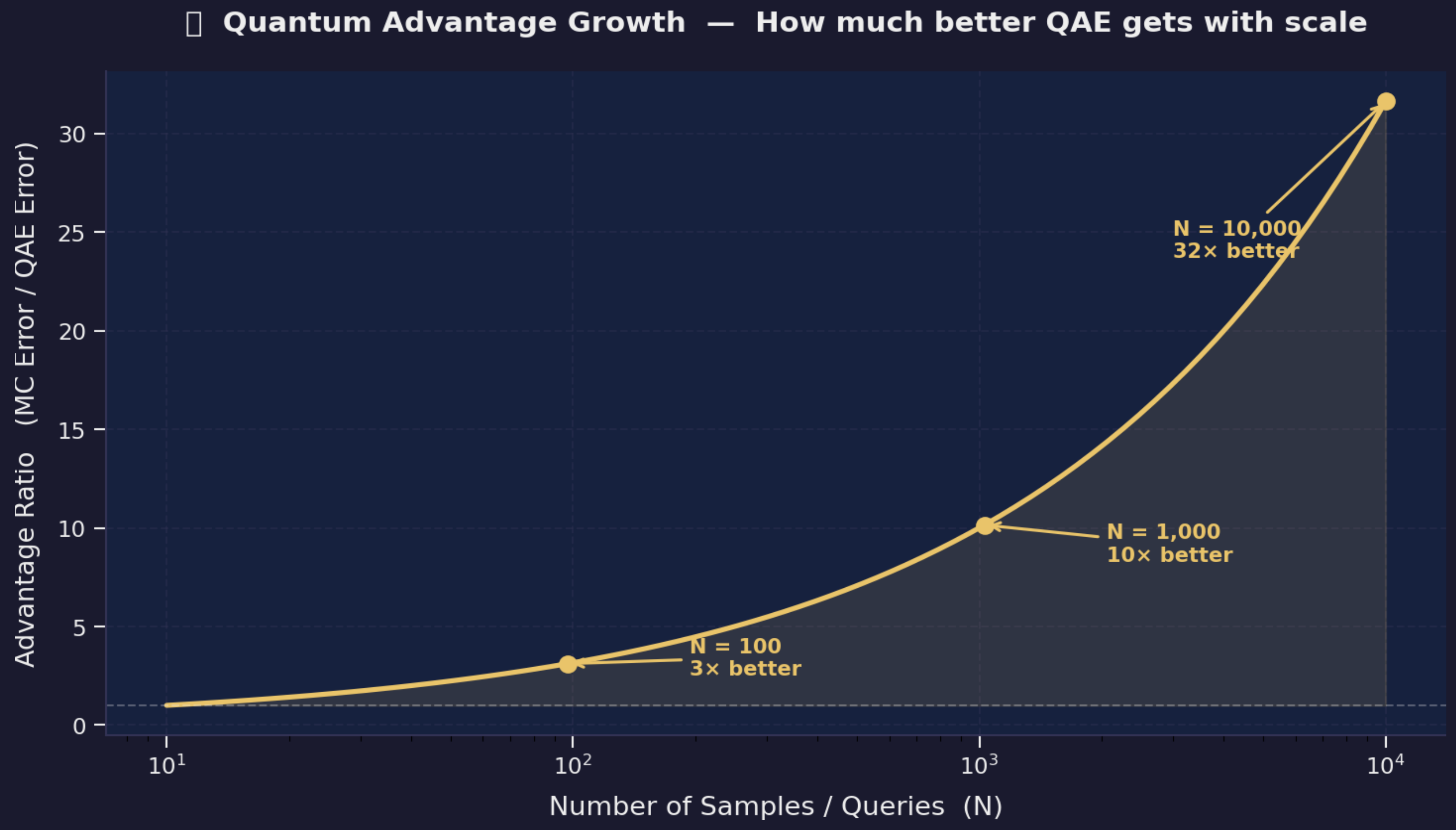
Key Insights

Heatmap shows profit across all price + uncertainty combinations

Higher uncertainty → higher profit variance → MC estimates become noisier

QAE's advantage is greatest where uncertainty is highest

🚀 Quantum Advantage Growth



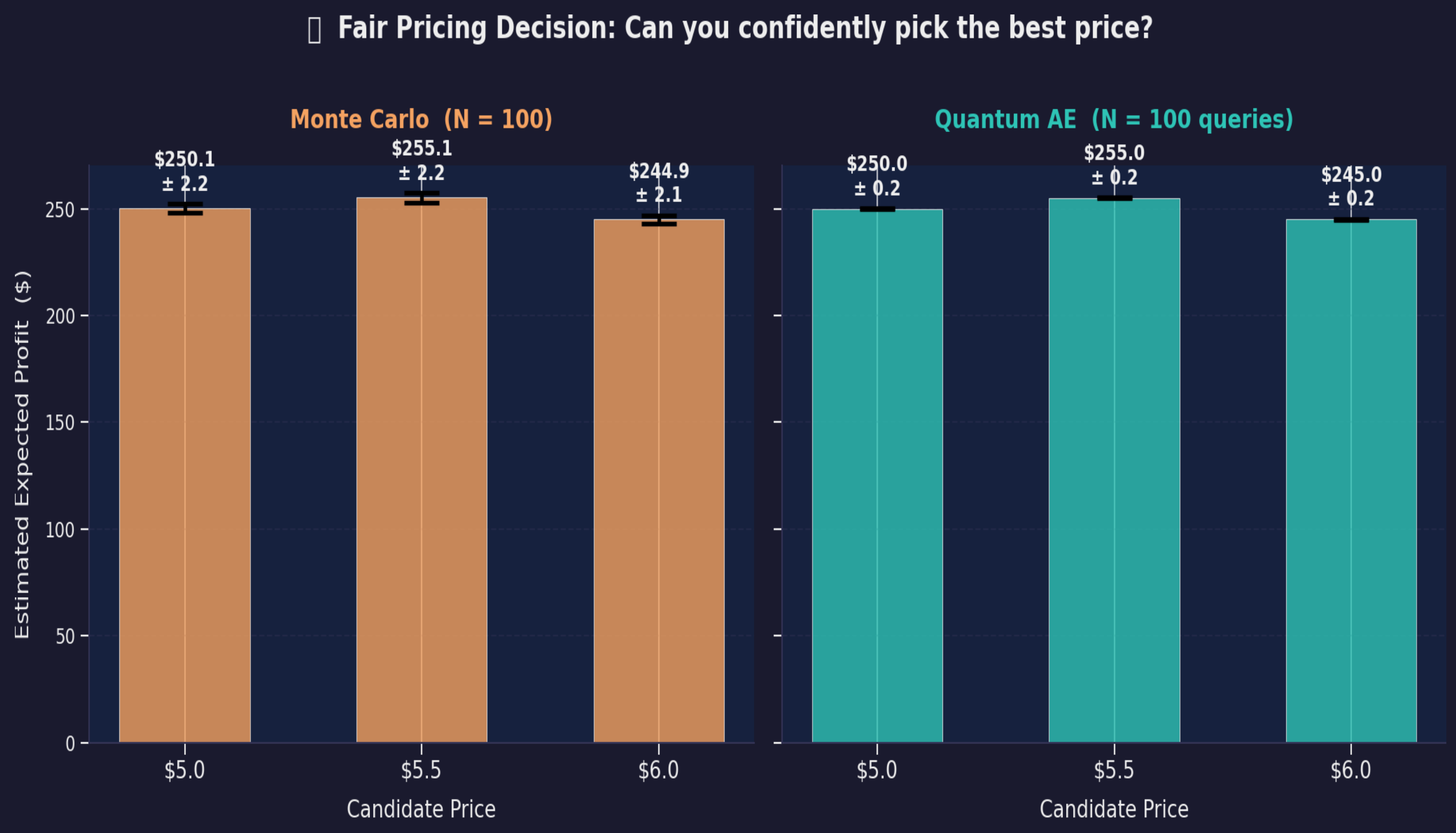
Key Insights

The ratio (MC error / QAE error) grows as √N

N=100: 3× better | N=1,000: 10× better | N=10,000: 31.6× better

The more you invest in computation, the more quantum pays off

🎯 Fair Pricing Decision: The Killer Chart



Key Insights

MC (left): error bars OVERLAP — ambiguous decision, cannot distinguish $5.0 from $5.5

QAE (right): clear separation — $5.50 is unambiguously the best choice

Same budget (N=100), dramatically different decision quality

⚖️ Full Comparison: Classical vs Quantum

Dimension	Monte Carlo	Quantum AE
What it computes	Sample average	Amplitude of quantum state
Core mechanism	Random sampling	Quantum interference
Error scaling	$O(1/\sqrt{N})$	$O(1/N)$
Cost for error ε	$O(1/\varepsilon^2)$	$O(1/\varepsilon)$
Error at $N = 100$	± $2.20	± $0.22
Error at $N = 10,000$	± $0.18	± $0.006
Advantage at N=10K	1× (baseline)	31.6×
Price stability	Noisy at low N	Stable even at low N
Decision quality	Ambiguous	Clear ✅

💡 The Fair Pricing Argument

Why Classical MC Falls Short

- **High estimation noise**
  Price appears optimal when it's not
- **Inconsistent decisions**
  Same analysis twice → different prices
- **Requires massive samples**
  Impractical for real-time pricing
- **Noise-driven decisions ≠ fair pricing**

Why Quantum AE Enables Fairer Pricing

- ✅ **10–30× lower noise**
  Price reflects true demand, not artifacts
- ✅ **Deterministic convergence**
  Same analysis → same decision
- ✅ **Linear scaling**
  Feasible for complex pricing models
- ✅ **Accurate estimation = fair pricing**

🧠 Executive Conclusion

"Quantum advantage in pricing is not about speed alone — it's about better decision quality under uncertainty."

Strategic Implications

- Near-term: Use classical MC with awareness of its limitations; build confidence intervals into pricing
- Medium-term: Explore hybrid classical-quantum approaches (variational QAE, noise-aware protocols)
- Long-term: Full QAE on fault-tolerant quantum hardware for high-precision, fair, real-time pricing

Value Proposition

For every pricing decision where $5 of profit difference matters, QAE can resolve the difference with 100× fewer evaluations. In high-frequency pricing, thin margins, and complex demand models — quantum advantage translates directly into better margins and fairer prices.

Thank You!

Questions & Discussion

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🍊 Fair pricing through quantum-enhanced estimation

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